

A study on the development of BIM-based energy efficient form design system for high-rise buildings in early design phase

YU SONGHUN

Architecture and Art Apartment
Beijing Jiaotong University
Beijing, China
19129010@bjtu.edu.cn

Wei Sun*

Architecture and Art Apartment
Beijing Jiaotong University
Beijing, China
sunwei@bjtu.edu.cn

PAK CHOL

Architecture and Art Apartment
Beijing Jiaotong University
Beijing, China
18129145@bjtu.edu.cn

ABSTRACT—At present, with the development of society and technological progress, super high-rise buildings are rapidly developing. At the same time, the energy consumption of high-rise buildings has become an important issue and the research on low-energy consumption building problems has always been an important topic in the field of super high-rise building design. Previously, most researches focused on the performance of equipment systems and building materials. However, in addition to study of efficient structural systems, more studies on the forms of high-rise buildings which are optimized in terms of energy efficiency are necessary.

The purpose of this research is to develop an energy-efficient form design system for BIM-based super high-rise buildings in the early design stage. A plug-in is developed to construct the super-tall building form in response to the environmental characteristics of different regions with access to the Application Program Interface (API) of the BIM tool and energy simulation tool. A plug-in generates architectural model by using Dynamo and Python. The simplified architectural model was converted to a gbXML based data model through interoperability technology. Then, the data model could generate an energy performance analysis model. Energy performance simulation is conducted for a high-rise building and the energy-efficient form suitable for the environmental characteristics of the region is decided by comparison.

KEYWORDS—BIM, Energy efficiency, High-rise building, early design phase, API(Application Program Interface), Dynamo, Python

I. INTRODUCTION

The current trend in global energy use raises serious questions about difficulties in supply, depletion of energy sources and serious environmental impacts [1]. In the face of uncertain global energy prospects, buildings should be designed and constructed to use less fossil fuel-based energy and have less environmental impact [2]. A quarter of the world's energy consumed is used for building operations, accounting for one-third of the world's carbon emissions. Building energy mainly uses fossil fuels, and the carbon emissions generated by using fossil fuels will increase the greenhouse gas content in the earth's atmosphere. Therefore, energy use and carbon emissions should be reduced by making buildings as energy efficient as possible.

High-rise buildings are a beneficial choice in densely populated areas where land is scarce, because they can be used effectively. However, since skyscrapers typically require more energy for construction and operation than low-rise buildings, the issue of energy-efficient design in the case of skyscrapers requires immediate attention [3]. Research on various plans to reduce energy consumption of skyscrapers in changing climates and design techniques to reduce greenhouse gases in cities is an important task to be addressed in the era of energy consumption.

The energy consumption of buildings varies in the defense and form of mass prescribed in the conceptual design stage, depending on the climate and external environment of the region. Because the existing energy performance evaluation of a building was generally evaluated through simulation after the design was completed, it is difficult to modify the shape of the building set in the conceptual design stage. Therefore, in terms of energy efficacy, from the beginning of the conceptual design stage, an integrated design strategy is needed to suit the external environment of the region.

The recent paradigm of the construction industry continues to make efforts for efficient energy management and utilization through the convergence of IT technologies. Building Information Modeling (BIM) is based on actual construction project-related information data, through which it is building a three-dimensional independent building model with digital technology. The building information model contains geometry, physics, and topological information of the building. Following the definition of BIM by NBIM, BIM is a shared knowledge resource [4]. The BIM design technique is based on a unified object information model, which enables collaboration from the conceptual design stage, and minimizes errors between tasks, making it a design method for large-scale projects such as skyscrapers. The BIM-based design is introduced in developed countries to build a sustainable urban environment with the advantages of solving environmental problems of skyscrapers at the conceptual design stage through reliable environmental simulation [5].

The purpose of this study is to develop a form design system for energy-efficient skyscrapers which can efficiently respond to various regional environments based on the results of reliable environmental simulation in order to seek the design strategy of high-rise buildings at the BIM early design stage in terms of

National key research and development project subject "Beijing-Zhangjiakou high-speed rail supporting visual design technology demonstration for the Winter Olympics" (No. 2020YFF0304106)

energy efficiency. We would like to develop an energy efficient form design plug-in using Revit API and Python [6].

II. DESIGN TECHNIQUE FOR BIM AND TALL BUILDINGS

A. BIM Design of Tall Building

1) Strategies for Eco-friendly Design of High-rise Buildings

a) Energy efficiency of BIM-based skyscrapers

The energy performance analysis program verifies the heat acquisition and loss of buildings by radiant heat from the sun,

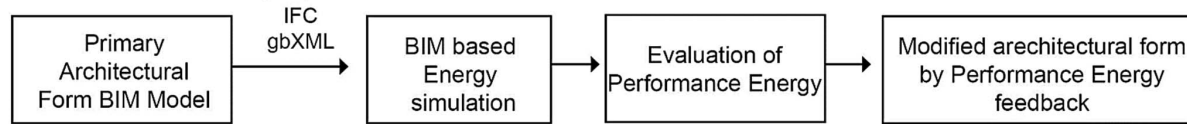


Figure 1. Design method using BIM-based energy performance analysis at the conceptual design stage.

b) Airflow of BIM-based Tall Buildings

The upper and lower floors of high-rise buildings differ in the pressure of air currents due to the difference in altitude and have a significant impact on the structural form. Wind loads directly affect the elevation of skyscrapers higher than surrounding buildings and are one of the important factors in choosing the form of buildings as well as structural characteristics. Since the characteristics of wind load vary greatly depending on the direction of the wind and the building's layout and shape, a proper design strategy should be established by predicting an air flow analysis in the conceptual design stage. Efficient skyscrapers are required to calculate the wind loads and derive them through wind tunnel which are experimental models at the stage of mass formation, but architectural designers are unable to access them due to lack of expertise. Thus, at the conceptual design stage, the architectural

and analyzes the load of mechanical facilities by calculation. After some parts of the design are completed, the energy performance is analyzed according to the demand for energy performance indicators for eco-friendly building certification.

Analysis of energy performance based on BIM requires an analytical model produced by BIM design tools. The analysis model is moved to a reliable energy performance analysis program, so it is necessary to supplement the omission of attribute information values of objects between files.

designer needs a methodology for a rapid review of the wind environment, in order to establish the form.

A simple simulation of BIM-based CFD software allows buildings to be predictable in direct or indirect wind environments and to reduce time spent selecting design alternatives.

Skyscrapers are higher than surrounding buildings, hence, they are directly affected by the wind. They, as well affect the direction of surrounding air currents. That being said, the surrounding buildings may have an adverse effect on the wind environment, thereby, when simulating, the models should be prepared and analyzed under the same conditions.

CFD tools for analyzing indoor and outdoor airflow are as follows and are linked to BIM programs through various file methods.

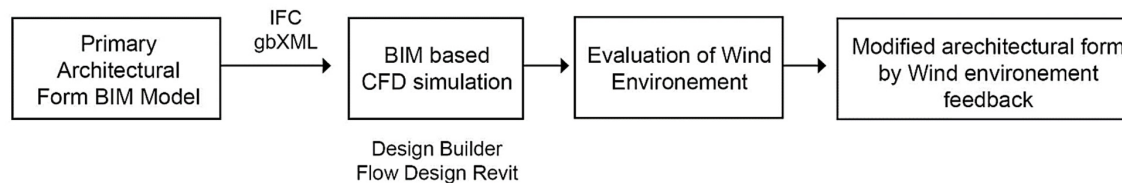


Figure 2. Design method using BIM-based CFD analysis at the conceptual design stage.

c) Structure of a BIM-based skyscraper

From the conceptual design stage, architectural designers should create a mass form while finding solutions through modeling and interpreting the BIM structure.

Irregular shaped buildings are subject to more loads than regular shaped buildings. Accordingly, development of external materials and new structural systems with structural efficiency are needed.

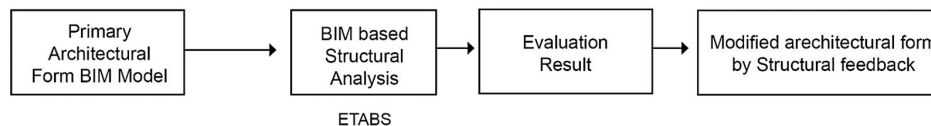


Figure 3. Mass design method using structural analysis based on BIM during conceptual design phase.

2) Methodology for Environmentally Friendly Design of Tall Buildings

The integrated design of high-rise buildings aims to carry out the design through various simulations to present the

optimal direction to operate the buildings with low energy from the conceptual design stage.

Design direction should be devised with structure and equipment experts at the conceptual design stage in order to

identify environmental factors that directly affect the design of high-rise buildings and present appropriate solutions based on reliable simulations.

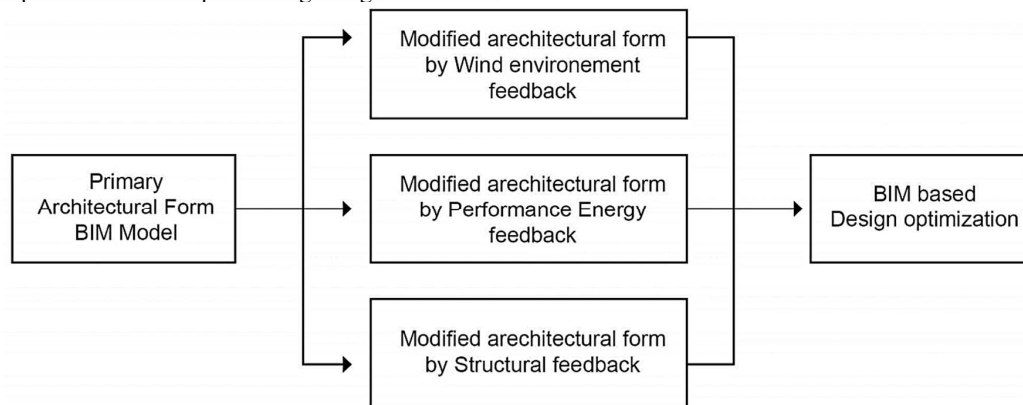


Figure 4. The BIM-based alternative to eco-friendly design at the conceptual design stage.

The picture below shows the structure of the eco-friendly integrated design of a building. When setting the form, cladding and layout direction of a building in the conceptual design stage, the external environmental factors around it should be analyzed to find alternatives. In order to derive the form of mass optimized for high-rise buildings based on BIM, alternative measures should be prepared through energy efficiency, wind load, and structural review.

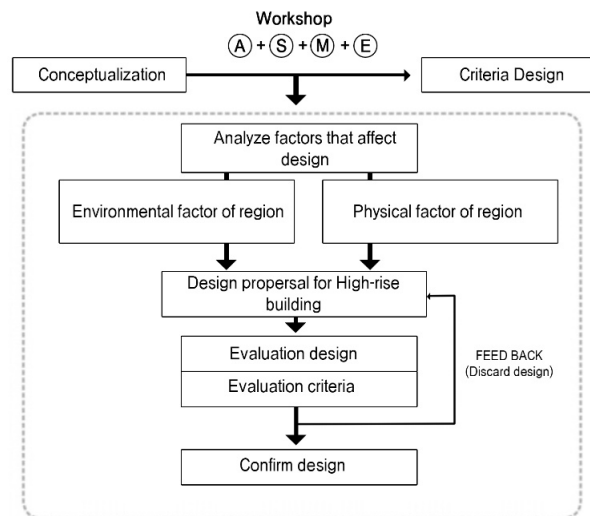


Figure 5. Integrated eco-friendly design structure.

B. Eco-friendly Design of Tall Buildings in Integrated Design Process

1) Composition and Role of Experts in Super High-rise Building Design

For skyscrapers, collaboration between architectural designers, structural engineers and equipment engineers is essential to improve the stability and environmental quality of the structural system from the conceptual design stage. The building's design determinants should be chosen with engineers and other experts, not by the architect's personal judgment.

Mir M. Ali (2006) divided the roles and design scope of architectural designers, structural engineers, and equipment engineers. Based on this, a design system chart of specialized areas for integrated design of high-rise buildings was prepared [7]. Architectural designers, as well as structural and facility engineers have a horizontal and mutual relationships. They design elements of each field which are not independent.

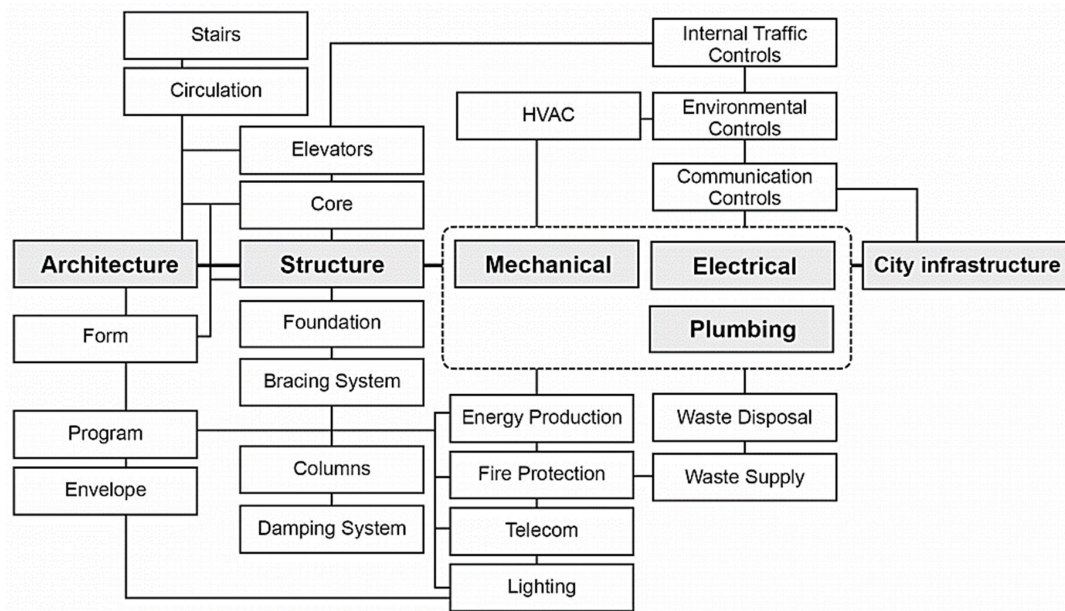


Figure 6. Schematic Design System for High-Rise Buildings.

The role of experts in the integrated design process of high-rise buildings should be clear and should be intervened at the right time to reduce time spent due to unnecessary work by mutual errors.

Beedle L divided the composition of the system into physical system, functional system, disaster system and building operation system [8].

2) Difference between integrated and basic design processes

High-rise buildings considering energy efficiency should establish an integrated design strategy optimized from the initial design stage, and design together with various experts based on various environmental simulations. However, the existing design process is linearly structured by dividing the areas of planning, designing and construction; into individual contracts.

In the initial stage, the designer first prepares the drawings and updates them through continuous cooperation with the structure and equipment. At this stage, the exchange of data is moved to independent files and documents, and numerous design errors and omissions consume the time of the project and cause economic loss. When the design is in its final stage, an energy analysis and evaluation will be conducted to measure more accurate energy consumption.

In the case of a super high-rise building, different from the general building design, the conceptual design stage determines a lot of design elements. In this type of projects, it is difficult to modify the conceptual design elements in terms of structure and system characteristics, even if the energy performance analysis results are low. For this reason, an integrated design process is suitable to minimize the cost spent in the operation stage by correcting errors in construction work early. This can be done by building reliable data that can be derived from conducting

collaborative screening based on communication with specialized fields from the conceptual design stage.

In order to understand the integrated design process for skyscrapers, we need to understand the composition of design participants first. Participants in the high-rise building design process can be explained in six categories by architectural designers, structural engineers, mechanical engineers, electrical engineers, constructors, and users. The following figures show the timing of participants' intervention in the design process.

The integrated design process method is based on various analyses and evaluations from one design stage to the next design stage, while the decision section may overlap and have design drawings with fewer errors in construction, resulting in smaller corrections in the construction part [9].

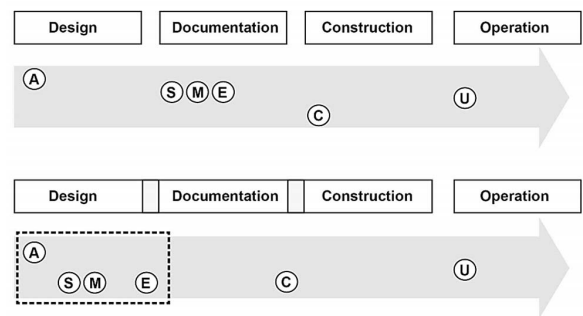


Figure 7. Existing design process (upper) and integrated design process (below)

III. ENERGY EFFICIENT DESIGN PROCESS FOR TALL BUILDINGS

A. Methodology for Form Generation of Tall Buildings in Conceptual Design Phase

In the conceptual design phase, an efficient alternative should be chosen by considering the factors that influence the form in creating mass forms. However, traditional working-level architectural designers focus on the architectural aspects and the aesthetic elements of architecture. Due to the

characteristics of existing skyscrapers, structural technology was important to review, and mass forms were developed in the conceptual design stage through structural analysis by gravity and air currents. The main element of form-making in the traditional design methodology of skyscrapers can be seen as the structural system. The design process for the structural system presented by the preceding researcher Mahjoub M^[10] is as follows.

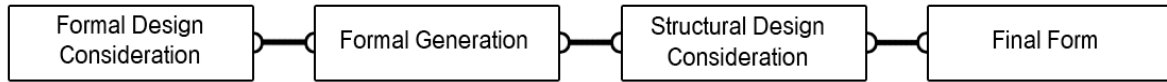


Figure 8. Existing methodology for mass formation of skyscrapers.

Although the form-generation methodology resulting from the review of the structural aspects can be derived in a stable form in the economics of subsidiary materials and the surrounding air currents, it is difficult to conclude that it is an efficient form-producing method in today's energy consumption era.

With the recent development of BIM technology, research and system development is being carried out to derive optimized forms and various virtual simulations of skyscrapers in the design stage. Mahjoub M, presented a design methodology incorporating energy efficiency, structural technology system and architectural form design for mass form generation using BIM program.

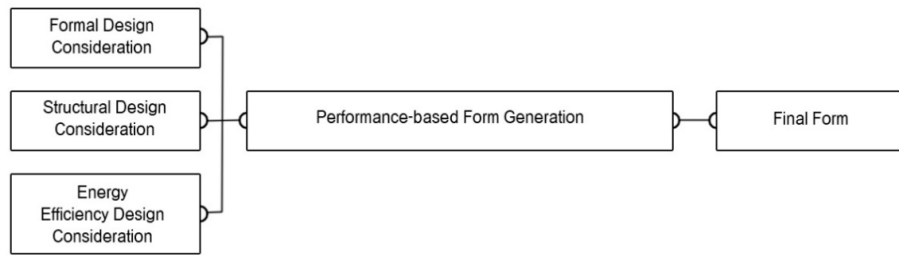


Figure 9. Proposal of Methodology for Mass Formation of High-rise Buildings by Integrated Method

Creating mass forms based on the review of energy efficiency and structural technology will enable optimized design for the surrounding environment.

B. Energy Efficient Design Process for Tall Buildings

1) Proposal Method Considering Energy Efficiency in Integrated Design Process

The existing energy performance analysis was performed by environmental experts based on the completed books, inputting necessary information data, and simulating them. Although the form of a building affects the performance of energy consumption, this is often not considered during the conceptual design phase. The interface of the energy performance analysis

program is a text-based input method, making it difficult for architectural designers to access it during the conceptual design phase, therefore there is no collaboration with environmental experts.

In the BIM-based integrated design process, the drawings are prepared using a single object information model, enabling appropriate energy performance analysis at each design stage, which can be utilized as a design methodology to generate mass forms with only the information values required during the conceptual design phase.

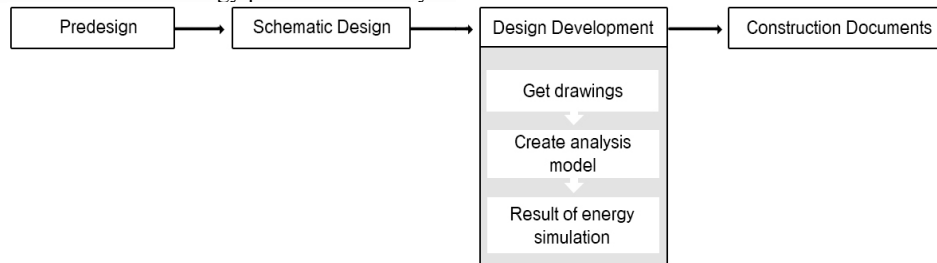


Figure 10. Figure 10: Energy performance analysis in the existing design process

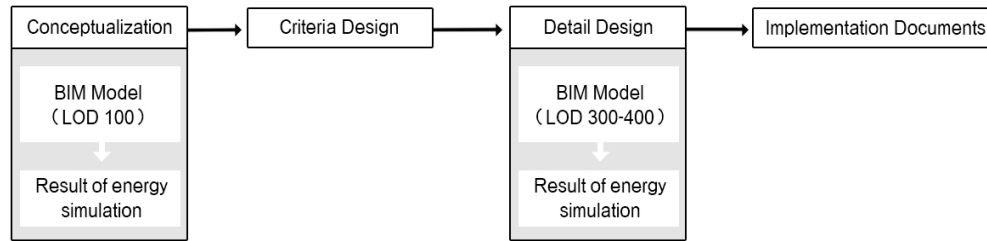


Figure 11. Figure 11: Energy performance analysis and utilization plan in the integrated design process.

A structured scenario is needed for the development of the formative process through the analysis of energy performance in the integrated design process. Problems with existing form generation methodologies should be identified and solutions to supplement them should be presented.

2) Energy-Efficient High-Rise Building Form Proposal Process Structure

In this study, the energy-efficient formative design process at the initial stage of BIM is sequentially divided into five categories.

This is a methodology and illustration for generating mass in the conceptual design phase.

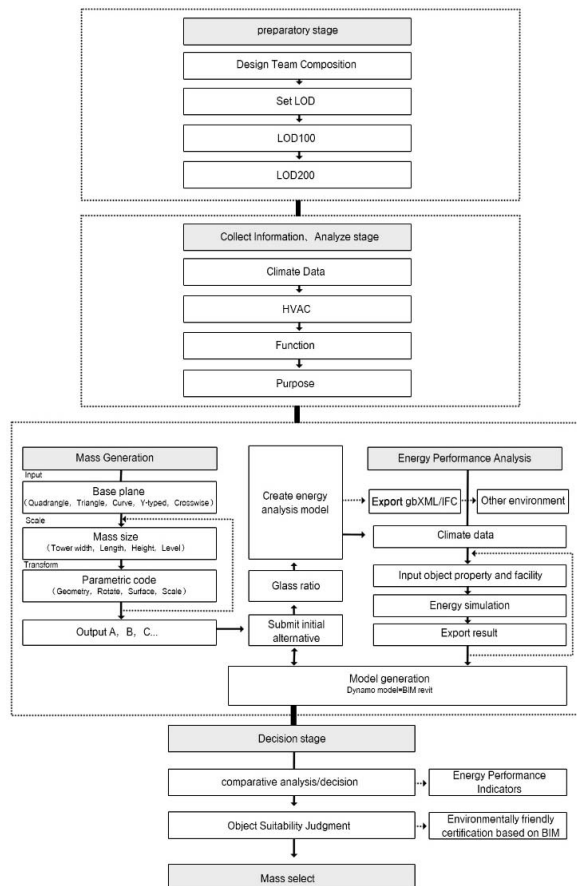


Figure 12. Energy-Efficient Form Design Process at the Early Design Phase of BIM.

IV. DEVELOPMENT OF ENERGY EFFICIENT FORM DESIGN SYSTEM

Energy efficient form generation systems aimed at deriving mass form alternatives are constructed as follows in the conceptual design phase.

1. Based on the analysis of energy performance, it aims to introduce alternatives in mass form using optimization techniques it is a means of promoting decision-making by architectural designers in the conceptual design stage.
2. The establishment of the system minimizes the loss of information between files by adopting the plug-in method of the BIM design program and increases the efficiency of the work by using a program that blends several specialized fields into an integrated design process. The BIM design program is based on Autodesk Revit and provides convenience to users through the development of architectural design-oriented interfaces.
3. It establishes a database that allows access to information necessary for the creation of an initial model, in order to facilitate the designers' work. DB adopts WEB method to obtain the latest information and uses existing skyscrapers as a basis for selecting the initial form model by providing the form classification and the corresponding energy performance analysis values.
4. It develops programming that automatically generates various forms based on the initial form model. Programming adopts visual graphic-oriented Dynamo and Python programming rather than text-based.
5. It analyzes energy performance based on appropriate regional information and reliable energy analysis engines in the formative form of programming. It adopts EnergyPlus engines that are widely used globally, and secures the old system that enables precise dynamic energy analysis after design as well as, the conceptual design stage of the integrated design process.
6. It provides indicators in order to find an alternative plan that is optimized through quantitative evaluation by comparing energy performance according to form.

A. Database Construction of High-rise Building Form Model

At the conceptual design stage, the architectural designer draws up the form of mass model and develops the shape of the building. Until mass models are created, architectural designers need to analyze various cases and make new types of attempts suitable for the region. If the form of a skyscraper is written in a database to fit the library classification system, architectural designers can more easily choose the form of an early generation model, which can increase the efficiency of their work.

B. Programming for Mass Generation

In this study, programming for Mass generation is developed as a platform available in Revit, based on Visual Programming Language (VPL) that allows architectural designers to easily change their intentions. Using Dynamo and Python, it is possible to quickly determine the mass form that meets the user's intention.

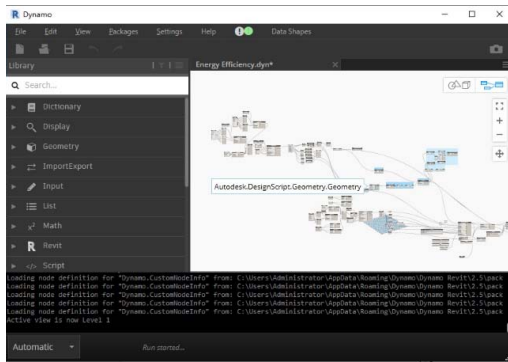


Figure 13. Dynamo Programming for Mass Generation.

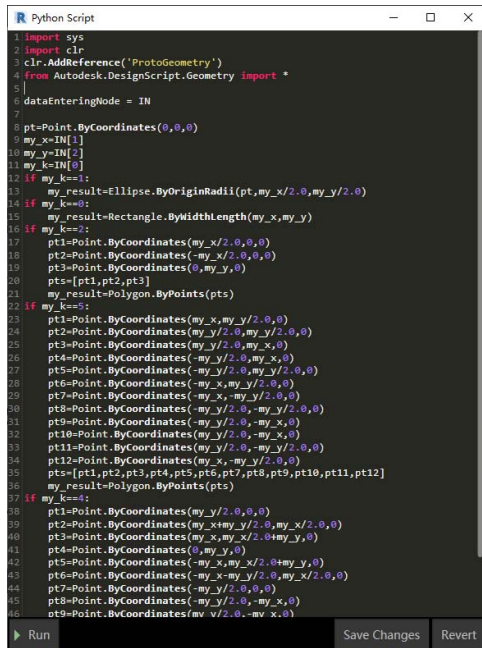


Figure 14. Python coding for Mass Generation

C. Establishment of Energy Performance Analysis System

Based on programming for form generation, the shape of the mass modified in the basic model is determined by the analysis of energy performance. In this study, energy analysis using Revit API is carried out and the result value and gbXML file are derived.

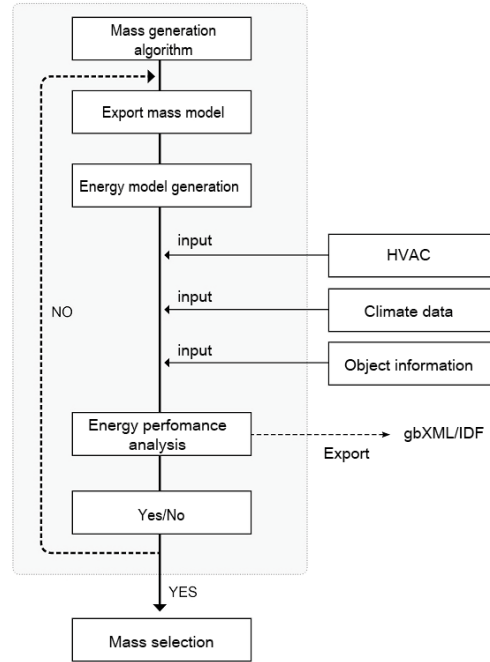


Figure 15. Structure of the energy performance analysis system.

D. Building a System User Interface

It is necessary to establish a user-centered interface of an energy-efficient form generation system for selecting optimal mass models by distinguishing the main factors of mass form generation considering energy efficiency and by entering fixed variables, such as the variables of the reference model and local climate that designers can choose.

E. Case study

We open Revit and click the icon shown below.

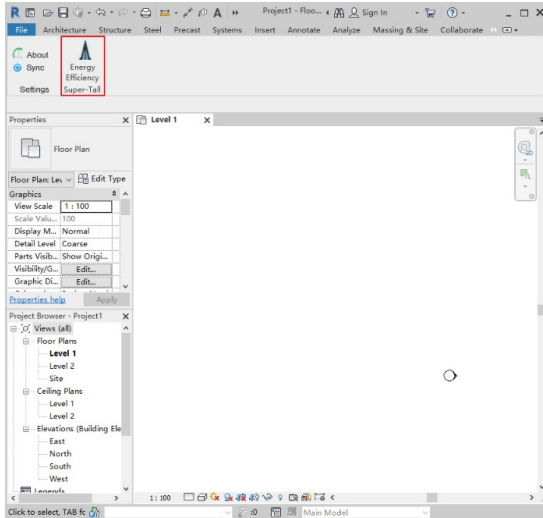


Figure 16. Revit's Plug-in "Super-Tall"

We consider a commercial building with a width of 30 meters and a width of 40 meters as an example. The bottom and upper planes are all square, and the number of floors is 70 stories, and the height is 3 meters. We set the parameters in the user interface.

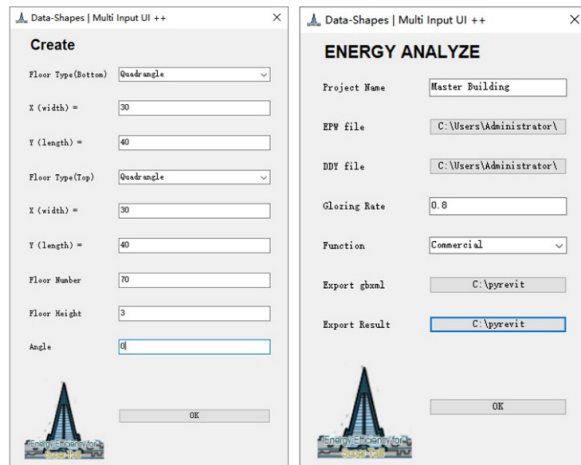


Figure 17. Parametric settings for energy performance analysis

The next energy performance analysis is carried out. The following figure is a result of the analysis.

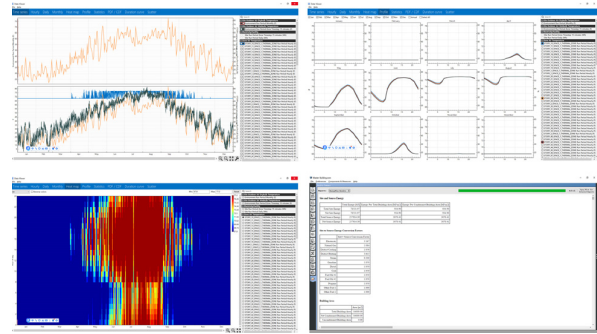


Figure 18. Analysis results.

V. CONCLUSION

This study sought a BIM-based design strategy for high-rise buildings in the conceptual design stage in order to find solutions to environmental problems caused by energy consumption of high-rise buildings. By proposing an energy-efficient formative design process for high-rise buildings during the initial BIM design phase and developing a plug-in form generation system, the architectural designer has to prepare a design technology that could reduce the time consumption for generating mass-type models and allow high-rise buildings, suitable for local environmental conditions, to determine their forms more quickly. We look forward to new developments in eco-friendly design based on BIM in the future.

ACKNOWLEDGMENTS

I sincerely thank Professor Sun Wei for her generosity and help during research. Although there were many difficulties during this period, I was able to finish the paper with the help of many teachers and friends including Professor Sun Wei. I will not be complacent, and I assume the responsibility of being modest and working hard from this day forth.

REFERENCES

- [1] Samuel E, Esther J, Richard A. (2016). Optimizing Energy Consumption In Building Design Using Building Information Model(BIM), Slovak Journal of Civil Engineering Vol. 24, No. 3, 19 - 28.
- [2] Rahman A, Narjes A. (2018). EMBODIED ENERGY OF BUILDINGS A Review of Data, Methods, Challenges, and Research Trends, Energy and Buildings(168), 225-235.
- [3] Maciej J. 2011. Energy Efficiency of Tall Buildings: Practical Methodology for Integrated Design, degree of Doctor of Philosophy in Architecture at The University of Auckland.
- [4] Chenyang Li, Lue Xie.(2017). Study on the Application of BIM Energy Analysis Technology in Architectural Design, International Conference on Construction and Real Estate Management.
- [5] Antony W. 2011. Tall Buildings: Search for a new typology, degree of Doctor of Philosophy, University of Nottingham.
- [6] Farzad J, Sephr M. (2020). An integrated BIM-LEED application to automate sustainable design assessment framework at the conceptual stage of building projects, Sustainable Cities and Society vol.53.
- [7] Ali M. 2006. Strategies for Integrated Design of Sustainable Tall Buildings,AIA Report on University Research, 14.
- [8] Beedle L. (2007). The Skyscraper and the City: Design, Technology, and Innovation,Lewiston, NY: Edwin Mellen Press.
- [9] Ajla Z. 2008. Knowledge-based model for integrated tall building design factors, degree of Doctor of Philosophy in Architecture in the Graduate College of the University of Illinois at Urbana-Champaign.

- [10] Mahjoub E. 2011.A Design Optimization Workflow for Tall Buildings Using Parametric Algorithm, CTBUH Research paper.